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CONSTANT CONDUCTANCE HEAT PIPES

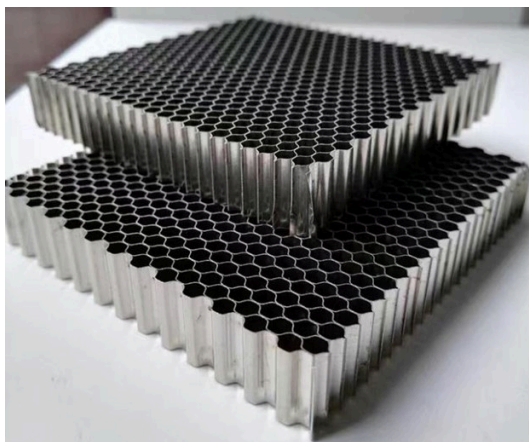
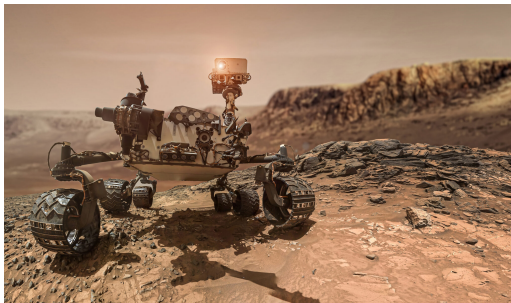
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THERMALLY ENHANCED HONEYCOMB PANELS FOR SPACECRAFT

Reliability is everything when it comes to sending any type of craft into the vastness of space. These spacecraft are highly specialized, very expensive pieces of equipment; there is emphasis on making sure that they work at top capacity through any situation.

Picture a planetary rover crawling the dusty surface of a planet for months on end. The days and nights are long and the landscape often brutal. The machine runs non-stop, controlling critical systems and using valuable power, all of which create waste heat that has the potential to damage onboard electronics. Thermal management systems are put in place to effectively dissipate and/or reuse that heat.



ACT has developed specialized **constant conductance heat pipe (CCHP)** embedded honeycomb panels, a type of thermally conductive composite radiator, consisting of a honeycomb structure with embedded constant conduction heat pipes. They can function as heat spreaders, radiators, or both. Honeycomb panels are ideal for a strong and rigid design in a lightweight package, but notoriously lack thermal conductance needed to effectively spread heat. The addition of CCHP's into the structure adds a minimal amount of mass, and a large amount of thermal spreading. A custom-designed CCHP-embedded honeycomb panel from ACT adds the thermal conductance exactly where it is needed most.

IDEAL APPLICATIONS



CCHP-embedded honeycomb panels are ideal for use on:

- Satellites
- Rovers
- Landers
- Space stations of any size

The honeycomb design provides a large surface area for heat dissipation and the constant conductance heat pipes embedded into the panels passively transfer heat from mounted components to another area of the spacecraft for dissipation. Typical CCHP-embedded radiator designs can produce a radiator efficiency of over 95% and keep temperature differences to just a few degrees Celsius.

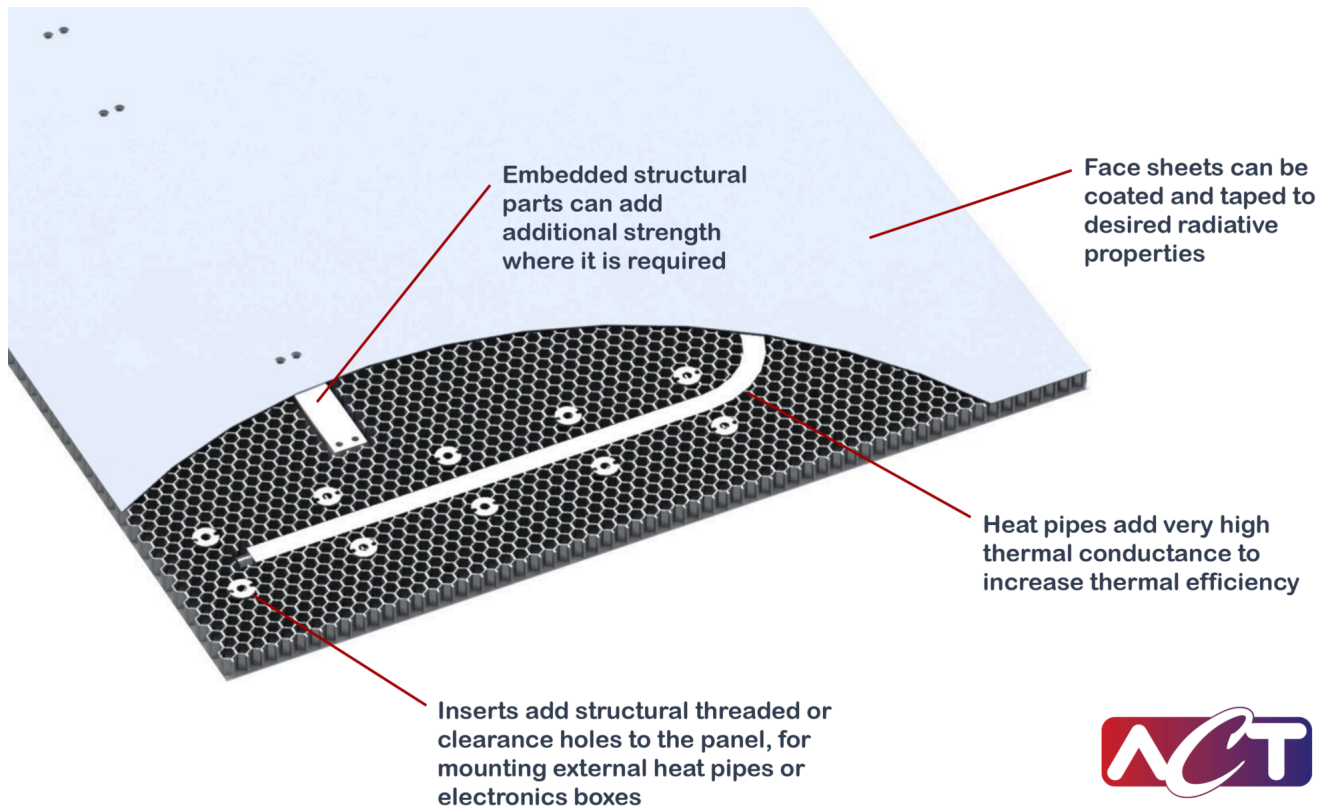
BENEFITS OF INCLUDING CCHPS IN HONEYCOMB PANELS

- The embedded CCHPs can also create an ideal interface location for externally mounted heat pipe technologies, all CCHP builds are custom so the team at ACT can add additional mounting locations for other pieces of equipment when necessary.
- They can ensure efficient thermal coupling between heat loads and the radiator source
- They can transfer large heat loads over long distances with minimal temperature drop. It is important to maintain a temperature gradient with minimal variant throughout the spacecraft, at the same time, it is key to move excess heat away from critical components and disperse them. Utilizing CCHP-embedded honeycomb panels is an easy way to passively control the thermal needs of your spacecraft without extra controls needed.



STRUCTURAL DESIGN





The unique honeycomb structure of these panels is both lighter and more thermally conductive than machined aluminum panels while remaining incredibly strong and able to support heavy electronic boxes. Inserts allow for threaded holes to have the strength of solid aluminum and parts made of other materials can be fully integrated into the panel for selected areas of very high strength tolerance. By designing a layout of internal thermal and structural components specific to the application's needs, the panel's functionality is significantly improved without increasing its size. ACT honeycomb panels with CCHPs can also be designed for thermal contact on both sides of the panel, allowing thermal connections between any components and mounting locations.

SPACE FLIGHT HERITAGE

ACT has over 70 million hours in orbit, making them uniquely suited to handle this level of specialty embedded heat pipe project. With thermal design and analysis as core competencies, and CCHPs as a core product line, ACT can design and manufacture custom honeycomb CCHP panels to reliably meet the demands of any application.

If you have any questions about thermal control systems for spacecraft or would like more information on ACT's capabilities in this area, please reach out to our team.



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